Graph Sketching: Radical Functions

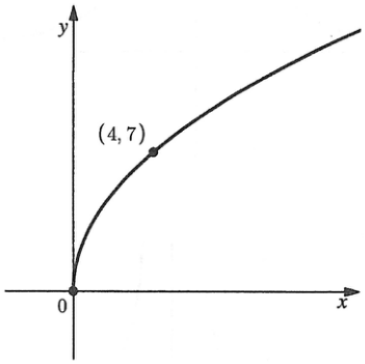
Questions 1–6

Question 1. Sketch the following graphs:

a) $y = \sqrt{x}$

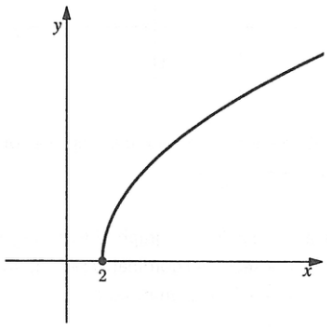
b) $y = -\sqrt{x}$

Question 2. Find the equation of the following graph in the form $y = k\sqrt{x}$.

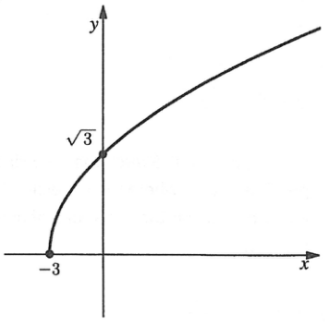


Question 3. Find the equation of each of the following graphs in the form $y = \sqrt{x - a}$.

a)

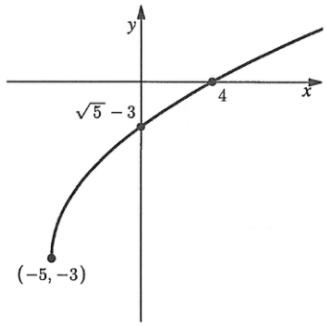


b)

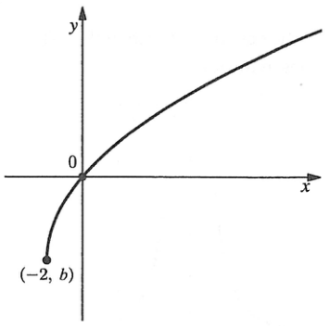


Question 4. Find the equation of each of the following graphs in the form $y = \sqrt{x - a} + b$.

a)



b)



Question 5. Sketch the following graphs:

a) $y = \sqrt{x-3} - 2$

b) $y = 4 - \sqrt{x+2}$

Question 6. The diagram below shows the graphs $y = 4 - \sqrt{x}$ and $y = \sqrt{x+4}$. Find the exact coordinates of the points of intersection.

